

PHY 517 / AST 443:

Observational Techniques in Astronomy

Lecture 6+:
Statistics part III

Bayesian statistic

Definitions:

- $P(A)$: probability of A happening
- $P(A, B)$: probability of both A & B jointly happening
- $P(A | B)$: probability of A happening **CONDITIONED** that B happens

Bayesian statistic

Bayes' theorem:

$$P(A, B) = P(A \mid B) * P(B)$$

$$P(A, B) = P(B \mid A) * P(A)$$

$$\Rightarrow P(A \mid B) = P(B \mid A) * P(A) / P(B)$$

Example: medical testing for rare disease

- Test is always positive when having the disease
 - $P(T+|H+) = 1$
 - $P(T-|H+) = 0$
- In one in 100,000 cases the test is falsely showing a positive result.
 - $P(T+|H-) = 1/100,000$
 - $P(T-|H-) = 1 - P(T+|H-)$
- *Q: You get a positive result. How likely is it that you actually have the disease? (Positive Predictive Value; PPV)*

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- *Prevalence of the disease is 1:1 Mio*
- *Q: You get a positive result. How likely is it that you actually have the disease? (Positive Predictive Value; PPV)*